Exercise for Computer Vision

Sheet 01

1 a] Distributive Property [Dicrete Time] :

We have $f[n] * [g[n] + h[n]] = [f[n] * g[n]] + [f[n] * h[n]]$

$\because$ $f[n]$: 2D array of input image

$g[n]$ and $h[n]$: two convolution kernels

$\because$ The definition for to convolution of two signals $x[n]$ and $y[n]$:

$$[x * y][n] = \sum_{k=-\infty}^{\infty} x[k] \cdot y[n-k] \quad \text{——— (1)}$$

$\because$ $[x*y][n]$: the output signal.

Now, LHS from (1), ~~let~~

$f[n] * [g[n] + h[n]]$, let $g[n] + h[n] = s[n]$ ——— (2)

$$\therefore f[n] * s[n] = \sum_{k=-\infty}^{\infty} f[k] s[n-k]$$, from (2) substituting back $s[n-k]$

$$= \sum_{k=-\infty}^{\infty} f[k] \cdot [g[n-k] + h[n-k]]$$

Using the distributive property of multiplication :

$$= \sum_{k=-\infty}^{\infty} [f[k] \cdot g[n-k] + f[k] \cdot h[n-k]]$$

Now, split the summation :

$$= \sum_{k=-\infty}^{\infty} f[k] g[n-k] + \sum_{k=-\infty}^{\infty} f[k] \cdot h[n-k] \quad \text{——— (3)}$$

Now comparing ③ with RHS:

we have

$$\sum_{k=-\infty}^{\infty} f[k]\, g[n-k] + \sum_{k=-\infty}^{\infty} f[k]\, h[n-k] \quad \text{--- ④}$$

$$[f[n] * g[n]] + [f[n] * h[n]] \quad \text{--- ⑤}$$

Using the definition ①, ④ and ⑤ are equal

$\therefore$ LHS = RHS

$$\therefore \boxed{f[n] * [g[n] + h[n]] = [f[n] * g[n]] + [f[n] * h[n]]}$$

b] Associative Property [Continuous Time]

We have $[f[t] * g[t]] * h[t]$ is equal to $f[t] * [g[t] * h[t]]$ --- ①

the definition of continuous time convolution:

$$x[t] * y[t] = \int_{-\infty}^{\infty} x[\tau]\, y[t-\tau]\, d\tau \quad \text{--- ②}$$

Now, let evaluate LHS: $[f[t] * g[t]] * h[t]$

$$\because f[t] * g[t] = \int_{-\infty}^{\infty} f[\tau_1]\, y[t-\tau_1]\, d\tau_1 \text{, from ②}$$

Let $f[t] * g[t] = y[t]$ --- ③

$\therefore$ LHS $= y[t] * h[t]$

$$\therefore y[t] * h[t] = \int_{-\infty}^{\infty} y[\tau_2]\, h[t-\tau_2]\, d\tau_2 \quad \text{--- ④}$$

$\because \tau_2$ is choosen for the outer convolution.

In ④, we have $y[\tau_2]$, $\therefore$ lets replace t with τ_2 in ③

$$\therefore y[\tau_2] = f[\tau_2] * g[\tau_2]$$

from ②, $y[\tau_2] = \int_{-\infty}^{\infty} f[\tau_1]\, g[\tau_2 - \tau_1] \cdot d\tau_1$ ——— ⑤

Substituting ⑤ in ④,

$$\therefore LHS = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f[\tau_1] \cdot g[\tau_2 - \tau_1] \cdot d\tau_1 \right] \cdot h[t - \tau_2] \cdot d\tau_2$$

$$\therefore [f[t] * g[t]] * h[t] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f[\tau_1] \cdot g[\tau_2 - \tau_1] \cdot h[t - \tau_2] \cdot d\tau_1 \cdot d\tau_2$$ ——— ⑥

Now, evaluating R.H.S : $f[t] * [g[t] * h[t]]$

Let start with inner convolution using integration variable τ_2

$$\therefore g[t] * h[t] = \int_{-\infty}^{\infty} g[\tau_2] \cdot h[t - \tau_2] \cdot d\tau_2$$ ——— ⑦

Now, let $z[t] = g[t] * h[t]$ — ⑧

$\therefore f[t] * z[t] = \int_{-\infty}^{\infty} f[\tau_1] \cdot z[t - \tau_1] \cdot d\tau_1$ ——— ⑨ , $\because$ using τ_1 for outer convolution.

Now from ⑧, $z[t - \tau_1] = g[t - \tau_1] * h[t - \tau_1]$,

using τ_2 as integration from ⑦

$$\therefore z[t - \tau_1] = g[\tau_2] \cdot h[[t - \tau_1] - \tau_2] \cdot d\tau_2$$ ——— ⑩

Now combining (9) (10),

$$f[t] * z[t] = \int_{-\infty}^{\infty} f[\tau_1] \cdot \left[\int_{-\infty}^{\infty} g[\tau_2] \cdot h[[t-\tau_1]-\tau_2] \cdot d\tau_2 \right] \cdot d\tau_1$$

$$\therefore RHS = f[t] * [g[t] * h[t]] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f[\tau_1] \cdot g[\tau_2] \cdot h[t-\tau_1-\tau_2] d\tau_2 \cdot d\tau_1 \quad \text{--- (11)}$$

Now Equating LHS, RHS :

$\because$ let $\lambda = \tau_2 - \tau_1$, $\therefore \tau_2 = \lambda + \tau_1$, $\therefore d\lambda = d\tau_2$ --- (12)

Substituting (12) in LHS

$$LHS = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f[\tau_1] g[\lambda] \cdot h[t-[\lambda+\tau_1]] d\tau_1 \cdot d\lambda$$

$$\therefore LHS = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f[\tau_1] \cdot g[\lambda] \cdot h[t-\tau_1-\lambda] \cdot d\tau_1 \cdot d\lambda$$

Swaping the order of integration and renaming λ to τ_2

$$LHS = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f[\tau_1] \cdot g[\tau_2] \cdot h[t-\tau_1-\tau_2] \cdot d\tau_2 \cdot d\tau_1 \quad \text{--- (13)}$$

from (11) and (13) , $\boxed{LHS = RHS}$